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Adversarial Attacks on Person Image Synthesis: Leveraging Pose and Appearance Controls in Denoising Diffusion Models

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ABSTRACT Advancements in pose and appearance control using denoising diffusion models have significantly transformed person image synthesis, enabling the generation of high-quality, diverse images aligned with specific input conditions. However, these models are vulnerable to adversarial attacks that exploit pose and appearance control mechanisms, compromising the reliability of the output. This article introduces a novel adversarial framework targeting pose-guided diffusion models, categorizing attacks into two groups: *appearance manipulation attacks* such as *Frequency Perturbations* (FP), *Ghosting* (GH), *Gaussian Aberrations* (GA), *Intensity Transformations* (IT) and *Chromatic Aberrations* (CA); and *pose control attacks*, specifically *Induced Misdirection* (IM). We demonstrate how these attacks manipulate either the source image or target pose to produce misaligned or degraded outputs. Our key contributions include a detailed taxonomy of attack types, a low-cost adversarial framework leveraging control inputs, and a comprehensive empirical analysis of model vulnerabilities. Experiments on the *Person Image Diffusion Model* (PIDM) model reveal that frequency-based attacks reduce *Structural Similarity Index Measure* (SSIM) by over 85% and increase the *Learned Perceptual Image Patch Similarity* (LPIPS) by over 90%, while Ghosting leads to an SSIM drop of more than 35% and an LPIPS increase of nearly 100%, indicating substantial visual and structural degradation. These results underscore the effectiveness and stealthiness of certain attacks. By exposing these vulnerabilities, our work provides a foundation for future research on adversarial robustness in pose-guided generative diffusion models.

INDEX TERMS Adversarial attacks, denoising diffusion models, pose-guided image synthesis.

I. INTRODUCTION

Diffusion models have emerge as a groundbreaking advancement in generative modeling, are renowned for their ability to synthesize high-quality data by reversing a carefully designed diffusion process [6]. These models iteratively refine random noise into coherent and structured outputs by leveraging feature relationships in the training dataset, ensuring precise alignment with desired results across hundreds to millions of steps. Their exceptional capability and adaptability establish diffusion models as a cornerstone of modern generative techniques. One of the most impactful applications of diffusion models lies in image synthesis, a domain that has been a focal point of generative modeling research for decades.

Among these, pose-guided person image synthesis stands out as a prominent area of interest due to its ability to generate realistic images of individuals conditioned on specific pose inputs [7], [18], [20], [24], [27], [34]. This process involves generating an image of a person in a specified pose by transforming a source image to align with the spatial structure of a target pose, ensuring both pose accuracy and photorealistic quality.

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Traditionally, image synthesis of pose-guided persons was dominated by *Generative Adversarial Networks* (GANs) [22], which aim to translate the style of a source image to a specific target pose by modeling complex pose transformations. Recent advances position diffusion models as a robust alternative to GANs for pose-guided synthesis. These models iteratively refine pose-conditioned inputs, generating high-quality pose-aligned outputs. Using diverse datasets with varied pose images, diffusion models textcolorbluedemonstrate remarkable effectiveness in producing photorealistic results, often exceeding GANs in conditional generation tasks [1], [11]. Initially designed for unconditional generation, diffusion models are now successfully adapted to conditional settings, setting new benchmarks in pose-guided image synthesis and advancing the state of the art.

Early approaches to pose-guided synthesis, such as pose-conditioned image generation [18], struggle with inconsistencies due to their dependence on the source and target pose. This is addressed by Esser et al. [8], who use a conditional U-Net architecture, whereas Siarohin et al. [28] enhance deformation with skip connections and affine transformations. While Zhu et al. [38] propose a pose-attentional adversarial network that suffers from feature blurring, Ren et al. [25] introduce GFLA, which uses global flow fields and occlusion masks for adaptive warping. Zhou et al. [36] improve high-resolution synthesis using a PatchMatch-based GRU approach, which improves upon methods such as CoCosNet [34]. Man et al. [21], Zhang et al. [33], and Zhou et al. [37] increase realism using parsing-based techniques, whereas the NTED framework [23] uses neural textures to generate a more controllable appearance.

Pose-guided person image synthesis using denoising diffusion models [11] introduces a flexible framework for modeling precise relationships between pose and appearance information in source and target images. However, this work highlights the potential adversarial vulnerabilities inherent in such models. Specifically, we demonstrate how manipulations to pose and appearance input can lead to altered model output, exposing critical risks. The inherent adaptability of diffusion models to inputs, combined with the ease of modifying pose and appearance conditions, lowers the barrier to adversarial attacks. We explore the adversarial exploitation of the model's input mechanisms. We term these attacks appearance manipulation and pose control attacks, directly reflecting the appearance and pose control inputs they are designed to subvert. Attackers can exploit these vulnerabilities to alter the output throughout the denoising process, posing significant risks to privacy and security. For example, adversarial perturbations to the synthesized medical images could lead to misdiagnosis or treatment errors in healthcare, while manipulated identity synthesis might bypass facial recognition systems in security applications.

In contrast to conventional adversarial attacks that primarily target classification tasks such as image classification or text sentiment analysis, injecting an adversarial attack on an image synthesis denoising diffusion model via appearance and pose control is distinct since the generated image carries the appearance of the source image while preserving the target pose. Hence, there is a need to establish a new definition specific to the intricacies of adversarial attacks on image synthesis denoising diffusion models.

To provide an in-depth analysis of this issue, we begin with a comprehensive review of pose-guided person image synthesis using denoising diffusion models, with a particular focus on the DeepFashion dataset [15]. Subsequently, we demonstrate the adversarial vulnerabilities inherent in these models, highlighting their functionality and potential for exploitation. The major contributions of our work are as follows:

- To the best of our knowledge, this study is the first to reveal that source image and target pose inputs in the PIDM model can be maliciously manipulated, providing a novel pathway for injecting adversarial attacks and significantly lowering the barrier to such exploitation.
- Through an in-depth analysis of pose-guided synthesis, we categorize adversarial attacks into two distinct categories. (1) *Attacks on the source image*, including FP, GA, GH, and IT. (2) *Attacks on the target pose*, focusing on pose misdirection.
- We conduct the first systematic analysis of diverse adversarial attacks (FP, GH, CA, IT, GA and IM) on PIDM, quantifying their impacts using SSIM/LPIPS metrics. Our results reveal that FP attacks are uniquely effective due to their perceptual stealth and ability to degrade texture fidelity, while GH and IM disrupt structural coherence. This analysis establishes a benchmark for evaluating the robustness of attacks in diffusion-based synthesis models.

The rest of the article is organized as follows: Section II reviews existing approaches and recent advancements in the field. Section III provides an overview of the foundational concepts related to pose-guided person image synthesis and denoising diffusion models. Section IV details the proposed approach, including the design and categorization of adversarial attacks. Section V presents a comprehensive analysis and evaluation of the existing attacks and an ablation study to examine the impact of disabling key components of PIDM on adversarial robustness. Section VI examines the implications of the findings and potential avenues for future research. Finally, Section VII summarizes the contributions and outlines the key takeaways of this study.

II. RELATED WORK

A. POSE-GUIDED IMAGE SYNTHESIS

Pose-guided image synthesis has advanced significantly with diffusion models, enabling applications in virtual try-ons and animation. Early GAN-based methods struggled with complex pose transformations, but diffusion models now dominate due to their iterative refinement process. PIDM [1] pioneers pose-guided synthesis using denoising diffusion, leveraging *texture diffusion blocks* (TDB) and *disentangled classifier-free* (DCF) guidance to align source textures with target poses.

Subsequent works like *Fusion embedding for Pose-guided Person Image Synthesis using a Diffusion Model* (FPDM) [13] and *Coarse-to-Fine Latent Diffusion* (CFLD) [16] simplify pose-texture alignment while improving fidelity. *Recurrent Pose Alignment and Gradient Guidance for Pose Guided Image Synthesis* (RePoseDM) [12] further enhances structural consistency via recurrent pose alignment. We focus on PIDM as the first diffusion framework explicitly modeling pose-appearance interactions, making it ideal for studying adversarial vulnerabilities.

Bhunia et al. [1] introduce PIDM, the first diffusion-based framework for pose-guided person image synthesis. PIDM [1], which builts on the denoising diffusion probabilistic model [11], represents a significant advancement in this domain. PIDM employs a DCF guiding mechanism and a novel texture diffusion module to align the style and pose of the generated image with user-provided inputs, ensuring high-quality outputs that meet user-defined conditions. PIDM introduces a robust framework that incorporates four essential components.

Noise Prediction Module (H_ϵ): This module extracts and encodes complex texture patterns from the *source image* (x_s). Multi-scale features are aggregated from different layers of H_ϵ, creating a detailed representation necessary for the denoising process.

Texture Encoder (H_N): The texture encoder incorporates TDB, which transfer rich multi-scale texture patterns smoothly from the source image to the target pose. This integration ensures distortion-free image synthesis by exploiting the correspondences between source and target appearances.

TDBs: Integrated into multiple layers of H_N, TDBs enable the efficient transfer of texture patterns across scales, contributing to the generation of high-quality images.

DCF Guidance: This mechanism [26] improves the conditioning signal of the target pose (x_p) and the source image (x_s) during the inference phase. It disentangles guidance signals, ensuring precise texture transfer and improved overall image quality.

Other notable approaches include the *Global Flow and Local Attention* (GFLA) framework by Ren et al. [25], which uses occlusion masks and flow fields to warp source images to match target poses. Zhou et al. [36] propose a hierarchy-based PatchMatch technique, assisted by GRU, to learn high-resolution correspondences, while CoCosNet [9], [34] improves texture transfer for finer image details. These methods demonstrate the evolution of pose-guided synthesis but remain vulnerable to adversarial manipulations.

B. ADVERSARIAL ATTACKS ON DIFFUSION MODELS

Despite their robustness, diffusion models are vulnerable to adversarial attacks, especially in conditional tasks like pose-guided synthesis. Recent research reveals three key vulnerabilities:

1) *Input Space Attacks:* Perturbation to conditional inputs can degrade output quality. For example, *AdvI2I* [31] shows that minor perturbations in image-to-image tasks cause significant deviations, while *AdvDiffuser* [4] synthesizes natural-looking adversarial examples.
2) *Latent Space Manipulation:* Attacks in latent space can alter outputs imperceptibly. Liu et al. [14] demonstrate identity manipulation in face synthesis, highlighting privacy risks.
3) *Transferable Attacks:* Chen et al. [3] craft perturbations transferable across tasks, raising concerns for deployable models.

Comparison and Contributions: Although prior work has examined adversarial attacks on conditional diffusion models, these efforts are largely limited to *single-modality conditioning* – such as text-to-image generation (e.g., perturbing text embeddings, as demonstrated in *AdvDiffuser* [4]) or generic image-to-image translation (e.g., injecting universal pixel-level noise, as explored in *AdvI2I* [31]). These approaches fail to address the unique challenges of pose-guided synthesis, where the model processes two distinct but independent inputs: *a source image governing appearance* and *pose keypoints controlling geometry*. Our work is the first to systematically investigate how adversarial perturbations can exploit this dual-input architecture. Unlike prior methods that treat all input uniformly, we design pathway-specific attacks: appearance manipulations that preserve perceptual plausibility, and pose misdirections that subtly misdirect the target pose. This approach reveals vulnerabilities inherent to the interplay between appearance and pose control — a dimension absent in single-input models.

III. PRELIMINARY

A. DIFFUSION MODELS

Diffusion models [11] are a class of probabilistic generative models that synthesize data by progressively transforming random noise into meaningful outputs. They achieve this by reversing a predefined process of adding noise to data, thereby learning the underlying data distribution. Unlike traditional generative models such as GANs, diffusion models operate through a step-by-step refinement process, allowing them to achieve higher fidelity and control over the generated content. The formal definitions of the forward diffusion process, backward generation, patameterization, and training objective are detailed in the fundamental work by Ho et al. [11], which provides the theoretical basis for the diffusion framework used in models like PIDM [1].

These models can be categorized into two types: unconditional and conditional diffusion models. Unconditional diffusion models generate outputs based solely on the learned data distribution, making them suitable for tasks where no additional guidance is needed. In contrast, conditional diffusion models incorporate extra parameters, such as textual descriptions, image references, or pose inputs, to direct the synthesis process and generate outputs aligned with user-defined conditions. This conditional capability has made diffusion models

particularly powerful for complex tasks like text-to-image generation, image inpainting, and pose-guided image synthesis.

B. IMAGE SYNTHESIS

Image synthesis refers to the process of generating new images based on specified input data, such as noise, text, or other visual or structural inputs. It plays a crucial role in applications spanning creative media, computer-aided design, virtual reality, and medical imaging. Advances in deep learning have dramatically improved the realism, diversity, and controllability of synthesized images, propelling the field forward. Image synthesis can be broadly categorized into different types based on the input modality or intended application:

- *Unconditional Image Synthesis [11]:* This approach generates images purely from a learned data distribution without external guidance.
- *Text-to-Image Synthesis [32]:*This task involves generating images based on natural language descriptions, enabling creative applications such as DALL· E and Stable Diffusion.
- *Pose-Guided Image Synthesis [23]:* This involves generating an image that preserves the visual appearance of a source image while adhering to the spatial configuration (pose) defined by a target input.
- *Style Transfer and Editing [10]:* In style transfer, the goal is to modify an image's appearance by applying stylistic elements from another image (e.g., transforming a photograph into a painting style). Image editing tasks focus on targeted modifications such as changing attributes or re-imaging scenes.
- *Image Inpainting and Super-Resolution [5]:* These techniques restore missing regions in images (inpainting) or enhance their resolution (super-resolution), making them essential for image restoration and enhancement applications.

Diffusion Models in Image Synthesis: Diffusion models have emerged as a state-of-the-art approach for image synthesis. Their capability to model complex data distributions makes them particularly effective for conditional synthesis tasks, including pose-guided person image synthesis [1], [12], [13], [16].

C. ADVERSARIAL ATTACKS IN GENERATIVE MODELS

Adversarial manipulations of generative models fall into distinct categories:

- *Adversarial perturbations:* Imperceptible input modifications crafted to fool models during inference (evasion attacks). In our context, these include frequency or pose manipulations that degrade output quality while remaining visually undetectable [29].
- *Backdoor Attacks:* Require poisoning the training data with visible triggers, causing the model to learn malicious behaviors that are activated during inference when the trigger is present.

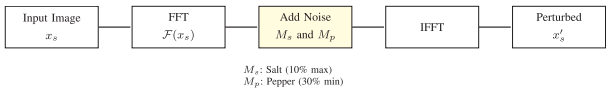


FIGURE 1. FP attack workflow.

While both threaten model integrity, they differ fundamentally in their perceptibility, implementation requirements (inference-only vs. training access), and defense strategies. Our synthesis, as they represent a more immediate and practical threat given the computational barriers to backdoor attack on large diffusion models.

IV. METHODOLOGY

The methodology of this study involves leveraging the principles of denoising diffusion models to synthesize person images, incorporating a multiphase process that systematically refines noisy inputs through iterative denoising steps. Our attacks target two distinct vulnerabilities. (1) Appearance manipulation perturbs the source image by degrading textures. (2) Pose control attack replaces target poses to misguide conditioning. While mechanistically independent, their combined use could simultaneously disrupt appearance and structure — a promising direction for stronger attacks.

A. ADVERSARIAL ATTACK BASED ON APPEARANCE MANIPULATION

This study systematically investigates five distinct types of adversarial attacks applied to the x_s for person image synthesis. Each attack manipulates specific aspects of the x_s, such as frequency components, pixel intensity, statistical noise, artifacts, or color channel alignment, to analyze their individual impact on the synthesized outputs. These attacks exploit different properties of the input image to introduce disruptions, either through noise, transformations, or artifacts, ultimately altering the generated outputs. While inspired by adversarial principles of input sensitivity and spatial transformations [17], [19], our appearance manipulation attacks are implemented using handcrafted perturbations without gradient optimization.

1) FP ATTACK

The implementation of the FP Attack modifies the frequency domain of the x_s to introduce specific perturbations as illustrated in Fig. 1. Let $x_s \in \mathbb{R}^{m\times n\times 3}$ be the source image matrix. The FFT(x_s), denoted as $\mathcal{F}(x_s)$ in Fig. 1, is applied to each color channel to obtain its frequency domain representation.

Binary masks (M_s) and (M_p) are generated as perturbations of salt and pepper noise, respectively, with a size of ($m \times n$). The values for the masks are set to 1 (salt) and 0 (pepper), based on random probabilities ($p_s = 0.1$) and ($p_p = 0.3$). These masks are applied to the frequency domain representation as: $(\text{FFT}(x_s)_{\text{salt}} = \text{FFT}(x_s) + M_s)$ and $(\text{FFT}(x_s)_{\text{pepper}} = \text{FFT}(x_s) - M_p)$ Finally, the inverse FFT (IFFT), $IFFT(\text{FFT}(x_s) + M_s)$ and $IFFT(\text{FFT}(x_s) - M_p)$, is applied to each channel to reconstruct the spatial domain perturbed channels, denoted as x_s(salt) and x_s(pepper). The

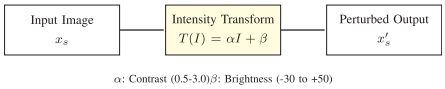


FIGURE 2. IT attack workflow.

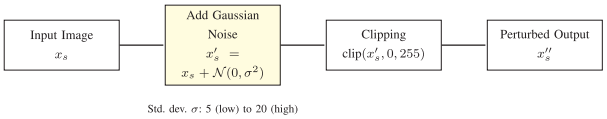


FIGURE 3. GA attack workflow.

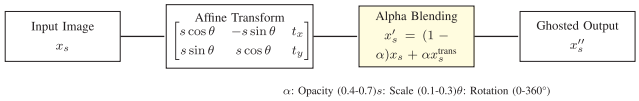


FIGURE 4. GH attack workflow.

final perturbed source image x_s(perturbed) (denoted as x'_s in Fig. 1) is formed by merging x_s(salt) and x_s(pepper) into an RGB image, expressed as:

$$x_s(\text{perturbed}) = \text{Merge}(x_s(\text{salt}), x_s(\text{pepper})). \tag{1}$$

In addition to salt-and-pepper noise, two other perturbation modes are introduced:

- Gaussian Noise: Random values are sampled from a normal distribution $\mathcal{N}(0, \sigma)$, where σ was proportional to the perturbation strength. These values are added directly to the frequency domain representation:

$$\text{FFT}(x_s)_{\text{gaussian}} = \text{FFT}(x_s) + \mathcal{N}(0, \sigma). \tag{2}$$

  This perturbation introduces smooth variations across the frequency domain components.
- Random Noise: Uniformly distributed random values are sampled from the range $[-R, R]$, with R determined by the perturbation strength. These values are added to the frequency domain as:

$$\text{FFT}(x_s)_{\text{random}} = \text{FFT}(x_s) + \text{Uniform}(-R, R). \tag{3}$$

  This method creates more disordered and diverse modifications to the frequency components.

The spatial domain perturbed channels are reconstructed using the IFFT for each type of perturbation. After processing the Red, Green, and Blue channels separately, the perturbed channels are combined to create the final RGB image.

All three perturbation types (salt-and-pepper, Gaussian, and random noise) are tested with varying strengths across 40 source images.

The perturbed images generated for each combination of perturbation type and strength are passed through the pose guided image synthesis pipeline to observe their impact. This systematic testing ensures a comprehensive evaluation of the robustness of the model against FP.

2) IT ATTACK

The second attack method applies IT $T(I)$ defined by the contrast factor (α) and brightness factor (β), which scales the contrast and shifts the brightness as illustrated in Fig. 2. The x_s(perturbed), denoted as x'_s in the Fig. 2, is obtained by applying $T(I)$ to the x_s:

$$T(I) = \text{ConvertScaleAbs}(I, \alpha, \beta). \tag{4}$$

$$x_s(\text{perturbed}) = T(I). \tag{5}$$

In order to thoroughly analyze the effects of intensity changes, various combinations of α and β factors are applied. These experiments offer a thorough examination of the effects of IT variations on tasks involving pose and appearance control.

3) GA ATTACK

The third attack method generates random Gaussian noise N with a specified standard deviation σ and adds this noise to the x_s to create an intermediate noisy image, x'_s as illustrated in Fig. 3. The pixel values of x'_s are then clipped to ensure they remain within the valid range of [0, 255], producing x_s(perturbed), which corresponds to x''_s in the Fig. 3:

$$x_s(\text{perturbed}) = I + N(0, \sigma^2). \tag{6}$$

The impact of rising noise levels on the robustness of the model is systematically assessed using these standard deviations.

4) GH ATTACK

The fourth attack method introduces ghosting effects to (x_s) by generating multiple ghost copies (n) with random transformations and blending them with the (x_s) using a specified opacity level. This method simulates ghosting artifacts, creating visual distractions as illustrated in Fig. 4.

The ghosted copy (x_s(ghosted_copy)), denoted as x_s^{trans} in Fig. 4 is created by applying an affine transformation (M) to the source image:

$$M = \begin{bmatrix} s \cdot \cos(\theta) & -s \cdot \sin(\theta) & \text{shift}_x \\ s \cdot \sin(\theta) & s \cdot \cos(\theta) & \text{shift}_y \end{bmatrix}. \tag{7}$$

The ghosting transformations involve applying random scaling s, rotation (θ), and shifts ($\text{shift}_x, \text{shift}_y$) within their respective specified ranges. The affine transformation is applied using:

$$x_s(\text{ghosted_copy}) = \text{warpAffine}(x_s(\text{original}), M, (\text{width}, \text{height})). \tag{8}$$

The ghosted image (x_s(ghosted)), denoted as x''_s in the Fig. 4, is created by blending each ghosted copy x_s(ghosted_copy) with the x_s using the specified opacity (denoted as α in the Fig. 4):

$$x_s(\text{ghosted}) = (1 - \text{opacity}) \times x_s + \text{opacity} \times x_s(\text{ghosted_copy}). \tag{9}$$

The GH attack is tested with various parameter combinations to analyze its effects (GH1-GH5).

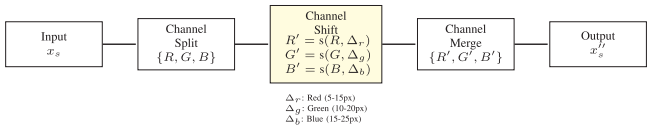


FIGURE 5. CA workflow showing RGB channel displacement.

TABLE 1. Impact of Adversarial Attack Against Appearance Manipulation on PIDM With the TDB and DCF Guidance on the DeepFashion Dataset

Methods	512×352 Resolution		256×176 Resolution	
	LPIPS ↓	SSIM ↑	LPIPS ↓	SSIM ↑
B1: PIDM with Both TDB and DCF Disabled	0.2643	0.7784	0.2374	0.7551
B2: PIDM with TDB Disabled	0.2618	0.7817	0.2339	0.7588
B3: PIDM with DCF Disabled	0.2660	0.7824	0.2393	0.7592
Fully Functional PIDM	**0.2550**	**0.7861**	**0.2284**	**0.7632**

5) CA ATTACK

The fifth attack introduces using displacement parameters $\Delta_r = 5 - 15$ pixels for red, $\Delta_g = 10 - 20$ pixels for green, and $\Delta_b = 15 - 25$ pixels for blue: $R' = \mathrm{s}(R, \Delta_r)$, $G' = \mathrm{s}(G, \Delta_g)$, $B' = \mathrm{s}(B, \Delta_b)$ where R, G, B are the individual color channels of the original image, and s represents the vertical shift operation. The resulting aberrated image (x_s(aberrated)), denoted as x_s'' in the Fig. 5, is as follows:

$$x_s(\text{aberrated}) = \text{Merge}(R', G', B'). \tag{10}$$

To investigate the influence of adversarial attacks, we conduct an empirical study in the experiment to determine which type of adversarial attack on appearance is most effective for PIDM. FP attacks exploit the spectral sensitivity of diffusion models by injecting perturbations in high-frequency bands, where human vision is less perceptive but model performance heavily relies on fine details. Formally, perturbing the $\mathcal{F}(x_s)$ with bounded noise Δ (e.g., $\|\Delta\|_2 \leq \epsilon$) disproportionately disrupts texture synthesis while remaining visually subtle. GH attacks degrade spatial coherence via affine transformations, creating conflicting cues that models struggle to resolve. FP attacks outperform spatial attacks (e.g., GA) due to their sparsity and alignment with human perceptual biases, as is evidenced by FP3's severe SSIM drop (-0.6864) versus GA3's milder impact (-0.1078) in Table 2.

B. ADVERSARIAL ATTACK BASED ON POSE CONTROL

Adversarial attacks targeting the pose control during the inference phase can lead to significant deviations from the intended pose generation. Instead of accurately replicating the prescribed target pose, the model produces outputs that deviate as a result of the adversarial attack. This results in the generation of a pose that appears random and unrelated to the input target pose.

1) IM

Let (x_p) denote the target pose, (x_p') the perturbed target pose, and (S_i) the poses generated from the sample. The perturbed target pose (x_p') is represented as:

$$x_p' = x_p + \epsilon, \tag{11}$$

where ϵ represents the adversarial perturbation.

TABLE 2. Quantitative Comparison of Denoising Diffusion Model for Person Image Synthesis (PIDM) Under Adversarial Attacks At 512 × 352 and 256 × 176 Resolutions on the DeepFashion Dataset

Attack Type	Settings/Parameters	512x352 Resolution		256x176 Resolution	
		LPIPS	SSIM	LPIPS	SSIM
None	Benign	0.2550	0.7861	0.2284	0.7632
FP1	Mode: Salt & Pepper, Strength: 5	0.4696	0.2980	0.4462	0.2868
FP2	Mode: Salt & Pepper, Strength: 10	0.5191	0.2072	0.4967	0.2011
FP3	Mode: Salt & Pepper, Strength: 20	**0.6354**	**0.0997**	**0.6191**	**0.0966**
FP4	Mode: Gaussian, Strength: 5	0.2589	0.7790	0.2314	0.7567
FP5	Mode: Gaussian, Strength: 10	0.2648	0.7759	0.2377	0.7527
FP6	Mode: Gaussian, Strength: 15	0.2626	0.7827	0.2358	0.7596
FP7	Mode: Gaussian, Strength: 30	0.2627	0.7734	0.2368	0.7499
FP8	Mode: Random, Strength: 5	0.2647	0.7784	0.2380	0.7553
FP9	Mode: Random, Strength: 10	0.2611	0.7789	0.2335	0.7558
FP10	Mode: Random, Strength: 20	0.2707	0.7724	0.2441	0.7494
FP11	Mode: Random, Strength: 50	0.2571	0.7813	0.2294	0.7589
IT1	Contrast: 0.5, Brightness: 0	0.3739	0.6152	0.3473	0.5924
IT2	Contrast: 3.0, Brightness: 0	0.4335	0.6426	0.4120	0.6181
IT3	Contrast: 1.0, Brightness: -30	0.2806	0.7362	0.2544	0.7154
IT4	Contrast: 1.0, Brightness: 50	0.3088	0.7527	0.2830	0.7281
IT5	Contrast: 2.0, Brightness: 50	0.4113	0.6685	0.3891	0.6439
IT6	Contrast: 1.5, Brightness: 30	0.3714	0.7028	0.3490	0.6779
GA1	Strength: 5	0.4196	0.6849	0.3886	0.6579
GA2	Strength: 10	0.4151	0.6816	0.3858	0.6539
GA3	Strength: 20	0.4246	0.6783	0.3941	0.6516
GH1	Num Ghosts: 3, Opacity: 0.5, Shift: 10	0.2839	0.7832	0.2525	0.7602
GH2	Num Ghosts: 10, Opacity: 0.7, Shift: 10	0.2852	0.7747	0.2554	0.7509
GH3	Num Ghosts: 50, Opacity: 0.5, Shift:10, Rot: 360, Scale:0.2	0.5120	0.5036	0.4884	0.4634
GH4	Num Ghosts: 20, Opacity: 0.4, Shift: 15, Rot: 180, Scale: 0.1	0.5245	0.4885	0.501	0.4455
GH5	Num Ghosts: 100, Opacity: 0.6, Shift: 20, Rot: 360, Scale: 0.3	0.5073	0.4956	0.4861	0.4547
CA1	Shift R: 5, G: 10, B: 15	0.3144	0.7616	0.2878	0.7410
CA2	Shift R: 15, G: 20, B: 25	0.3170	0.7556	0.2904	0.7352
IM	Random	0.5186	0.5783	0.5381	0.6063

The pose generation process involves randomly selecting a pose from the sample, denoted as (S_r). When adversarial perturbations are applied, the generated pose (S_g) is defined as:

$$S_g = S_r + \epsilon. \tag{12}$$

These deviations demonstrate how adversarial perturbations disrupt the pose generation process, causing the model to produce unexpected outputs that deviate from the intended target pose. Ensuring that the generated pose remains within a specified range before visualization is essential for accurately capturing the impact of adversarial attacks on pose-guided image synthesis.

The objective of these attacks is to expose vulnerabilities in denoising diffusion models, particularly for person image synthesis. By simulating potential security threats, this study highlights the risks these models may face in real-world applications and underscores the importance of developing robust defenses.

V. RESULTS

A. EXPERIMENTAL SETUP

We adopt the pioneering approach presented in [1], which employs a diffusion-based framework for pose-guided person image synthesis. This method integrates a TDB to model the interconnection between appearance and pose, along with a DCF guidance mechanism embedded in the sampling procedure. These features facilitate the evaluation of various adversarial attack scenarios targeting pose and appearance control.

For our experiments, we use the DeepFashion In-shop Clothes Retrieval Benchmark [15], a large-scale dataset comprising 52,712 high-resolution images of fashion models. This dataset is split into training and testing subsets, consisting of 101,966 and 8,570 pairings, respectively. The images include

a diverse range of clothing styles, poses, and appearances, providing a robust foundation for pose-guided synthesis evaluations. To extract pose information, we utilize OpenPose [2], a widely-used tool for obtaining skeleton-based pose representations from images.

Our experimental setup evaluates the robustness of the target model under different adversarial attack strategies. Specifically, we test the model's performance under adversarial perturbations applied to the source image and incorrect mapping attacks targeting the pose. These experiments aim to analyze the effectiveness of adversarial attacks while maintaining the integrity of the generated outputs.

B. EVALUATION METRICS

To evaluate the performance of the adversarial attack, we utilize two widely adopted metrics: SSIM [30] and LPIPS [35]. SSIM measures the similarity between images at the pixel level, focusing on luminance, contrast, and structural information, while LPIPS calculates the perceptual distance between generated and ground truth images, capturing higher-level discrepancies in visual semantics and perceptual quality. We employ a pretrained PIDM [1] to analyze whether adversarial attacks modify the semantic concepts in the generated images. These metrics [30], [35] have been extensively used in prior work on pose-guided image synthesis, including models developed before PIDM, making them a well-established standard for evaluation in this domain. By leveraging SSIM and LPIPS, we ensure a comprehensive assessment of the adversarial attack's impact on both pixel-level reconstruction accuracy and perceptual quality, enabling a direct comparison of the Adversarial PIDM's performance with other existing methods.

C. QUALITATIVE VISUAL RESULTS

Figs. 6 through 11 present qualitative comparisons of PIDM model outputs under adversarial attacks targeting pose and appearance control, respectively.

Pose Control Attacks: Fig. 11 compares the model outputs in three settings: Ground Truth (GT), PIDM, and IM. The results highlight significant disruptions in pose alignment and spatial accuracy caused by the attack, demonstrating how the IM successfully compromises the geometric integrity of the synthesized output. In particular, a random or unintended pose is chosen as a result of the adversarial perturbations misdirecting the pose control mechanism.

Appearance Manipulation Attacks: Figs. 6, 7, 8, 9, and 10 compare the benign PIDM outputs with adversarial PIDM outputs for five different attack types: FP, IT, GH, CA, GA.

Fig. 6 illustrates the effects of FP attacks. As attack strength increases (FP1 to FP3), high-frequency artifacts are introduced that gradually distort texturing and structural integrity. The adversarial outputs show a progressive degradation in visual fidelity, with an increasing loss of fine-grained details in complex scenes, with FP3 causing severe degradation. Similarly, contrast and brightness are impacted by IT Attacks, as illustrated in Fig. 7. These attacks introduce unique degradations with each IT, significantly affecting texture consistency, contrast, and brightness. The results show observable distortion in fine details, highlights, and shadows, with visual performance deteriorating markedly as the attack parameters deviate further from benign conditions. While slight changes



FIGURE 6. Visual comparison of PIDM outputs under FP Attacks. Rows represent: (1) inputs (target poses and source image), (2) ground truth (GT) outputs, (3) benign PIDM outputs, and (4-6) adversarial PIDM outputs for perturbation strengths FP1-FP3.



FIGURE 7. Visual comparison of PIDM outputs under IT Attacks. Rows represent, from top to bottom: inputs (target poses and source image), ground truth (GT) outputs, benign PIDM outputs, and adversarial PIDM outputs under different levels of IT, ranging from IT1 to IT5.

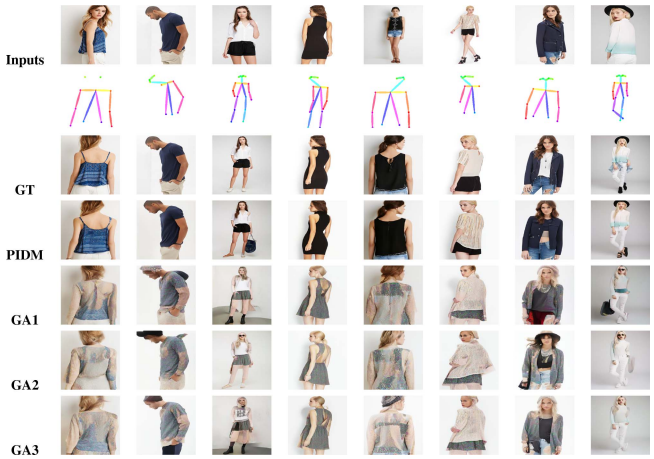


FIGURE 8. Visual comparison of PIDM outputs under **GA Attacks**. Rows represent, from top to bottom: inputs (target poses and source image), ground truth (GT) outputs, benign PIDM outputs, and adversarial PIDM outputs under different levels of GA, ranging from GA1 to GA3.

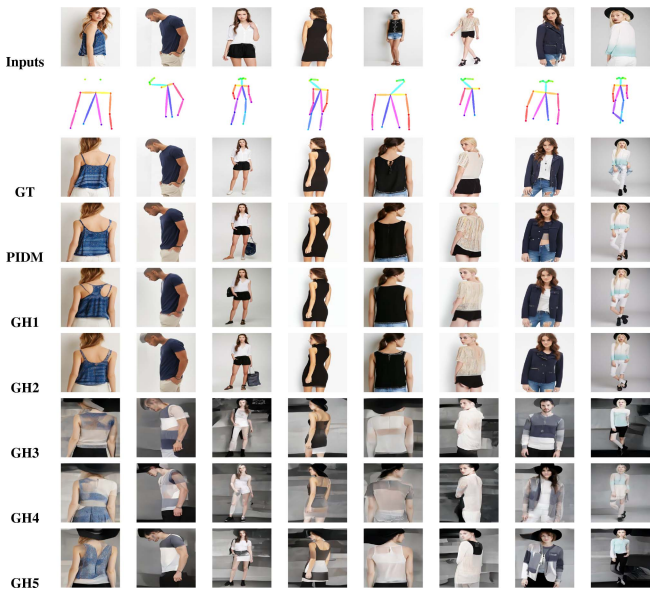


FIGURE 9. Visual comparison of PIDM outputs under **GH Attacks**. Rows represent, from top to bottom: inputs (target poses and source image), ground truth (GT) outputs, benign PIDM outputs, and adversarial PIDM outputs under different levels of GH, ranging from GH1 to GH5.

(IT1) result in subtle degradation, stronger intensities (IT4, IT5) cause overexposure, darkening, and uneven texturing.

Key characteristics are obscured by overlapping patterns created by GH attacks, which are shown in Fig. 9. These attacks cause duplicated or misplaced patterns, resulting in visual artifacts that severely degrade texture consistency and disrupt spatial alignment. The quality of images is greatly reduced by stronger attacks (GH3-GH5), as a higher number of ghosts and increased opacity leads to more pronounced aberrations, including clearly misaligned features and ghostly overlays. Fig. 10 illustrates the effect of CA attacks. These



FIGURE 10. Visual comparison of PIDM outputs under **CA Attacks**. Rows represent, from top to bottom: inputs (target poses and source image), ground truth (GT) outputs, benign PIDM outputs, and adversarial PIDM outputs generated with varying strengths of CA, ranging from CA1 to CA2.

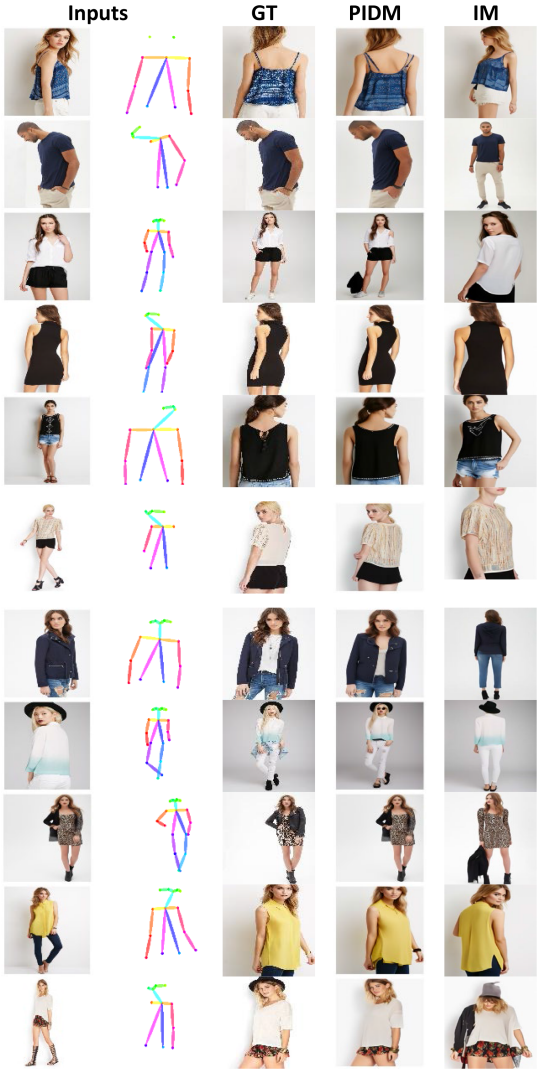


FIGURE 11. Visual comparison under **Pose Control Adversarial Attack**. Columns from left to right: input images, target poses, ground truth (GT) outputs, standard PIDM outputs, and IM adversarial outputs.

attacks cause prominent color fringing and spatial distortions, particularly noticeable at the edges of fine details and objects. The results demonstrate a clear increase in degradations to both spatial alignment and color consistency as the intensity of the channel displacement grows from CA1 to CA2, which significantly disrupts the overall visual fidelity of the synthesized images.

Fig. 8 illustrates GA attacks. These attacks introduce varying degrees of blurring and noise, which directly degrade texture consistency and edge sharpness. The adversarial outputs demonstrate a clear increasing loss of fine details and overall degradation in visual fidelity as the strength of the noise intensifies from GA1 to GA3, with higher intensities (GA3) causing severe deterioration of features.

GA and FP significantly disrupt texture and sharpness, while IT affects the balance of contrast and brightness. The overlapping distortions caused by GH obscure content, while CA disrupts color fidelity and spatial alignment. Robustness against a variety of adversarial perturbation is essential, as each attack highlights a distinct vulnerability in the model.

D. QUANTITATIVE METRIC EVALUATION

The quantitative results provide numerical evidence of the impact of adversarial attacks on fully functional PIDM performance, evaluated using SSIM and LPIPS metrics. In the context of evaluating adversarial attacks, the objective is to degrade the model's output. Therefore, a successful attack is indicated by a decrease in the SSIM score (signifying a loss of structural similarity to the ground truth) and an increase in the LPIPS score (signifying a greater perceptual distance from the ground truth).

We compute percentage changes in SSIM and LPIPS relative to the benign baseline to quantify the impact of adversarial attacks as shown in Table 2. For the highest-strength FP3 attack, SSIM drops from 0.7861 to 0.0997 at 512x352 resolution — a reduction of 87.3%, while LPIPS rises from 0.2550 to 0.6354 — an increase of 149.2%. GH3 attack shows a 35.9% drop in SSIM (0.7861 $\rightarrow$ 0.5036) and a 100.7% increase in LPIPS (0.2550 $\rightarrow$ 0.5120). IM attack leads to 26.4% SSIM decrease and 103.4% LPIPS increase. These metrics confirm that appearance and pose based adversarial attacks significantly degrade the model's structural and perceptual fidelity.

At lower strengths, other attack types demonstrate comparatively lesser impact, as evidenced by LPIPS and SSIM in Table 2. However, these attacks also introduce noticeable degradations as attack strength increases. Overall, these adversarial attacks deteriorate visual fidelity and structural consistency, with FP (Salt and Pepper) and GH attacks being the most detrimental, according to the LPIPS and SSIM metrics.

E. ABLATION STUDY

We evaluated three distinct configurations of PIDM on the DeepFashion dataset by disabling key components. The PIDM model with both TDB and DCF disabled is denoted as B1, the PIDM model with only TDB disabled is denoted as B2, and the PIDM model with only DCF disabled is denoted as B3 in Tables 3 and 4. The impact of TDB and DCF is evaluated by comparing their performance against fully functional PIDM using LPIPS and SSIM metrics across various attacks as presented in Table 2.

TABLE 3. Quantitative Comparison of PIDM Configurations (B1: Both TDB & DCF Disabled, B2: TDB Disabled, B3: DCF Disabled) Under Adversarial Attacks Against Appearance Manipulation At 512 × 352 Resolutions

Attack Type	Settings/Parameters	B1		B2		B3	
		LPIPS	SSIM	LPIPS	SSIM	LPIPS	SSIM
None	Benign	0.2643	0.7784	0.2618	0.7817	0.2660	0.7824
FP1	Mode: Salt & Pepper, Strength: 5	0.4656	0.3512	0.4806	0.3174	0.4705	0.3512
FP2	Mode: Salt & Pepper, Strength: 10	0.5699	0.1664	0.5701	0.1547	0.5530	0.2190
FP3	Mode: Salt & Pepper, Strength: 20	**0.6226**	**0.0987**	**0.6496**	**0.0800**	**0.6201**	**0.0925**
FP4	Mode: Gaussian, Strength: 5	0.2651	0.7771	0.2581	0.7811	0.2601	0.7781
FP5	Mode: Gaussian, Strength: 10	0.2687	0.7737	0.2607	0.7817	0.2635	0.7807
FP6	Mode: Gaussian, Strength: 15	0.2665	0.7761	0.2622	0.7780	0.2555	0.7826
FP7	Mode: Gaussian, Strength: 30	0.2593	0.7765	0.2609	0.7707	0.2636	0.7790
FP8	Mode: Random, Strength: 5	0.2602	0.7798	0.2565	0.7792	0.2569	0.7871
FP9	Mode: Random, Strength: 10	0.2628	0.7751	0.2627	0.7745	0.2631	0.7801
FP10	Mode: Random, Strength: 20	0.2711	0.7702	0.2595	0.7790	0.2604	0.7769
FP11	Mode: Random, Strength: 50	0.2613	0.7779	0.2551	0.7814	0.2643	0.7770
IT1	Contrast: 0.5, Brightness: 0	0.3704	0.6228	0.3778	0.6061	0.3686	0.6189
IT2	Contrast: 3.0, Brightness: 0	0.4283	0.6477	0.4310	0.6408	0.4335	0.6443
IT3	Contrast: 1.0, Brightness: -30	0.2808	0.7360	0.2876	0.7315	0.2794	0.7405
IT4	Contrast: 1.0, Brightness: 50	0.3046	0.7535	0.3120	0.7490	0.3098	0.7538
IT5	Contrast: 2.0, Brightness: 50	0.4102	0.6581	0.4074	0.6673	0.4122	0.6616
IT6	Contrast: 1.5, Brightness: 30	0.3630	0.7076	0.3645	0.7029	0.3662	0.7041
GA1	Strength: 5	0.4114	0.6866	0.4116	0.6847	0.4117	0.6783
GA2	Strength: 10	0.4172	0.6820	0.4217	0.6789	0.4248	0.6820
GA3	Strength: 20	0.4233	0.6853	0.4231	0.6831	0.4287	0.6657
GH1	Num Ghosts: 3, Opacity: 0.5, Shift: 10	0.2806	0.7807	0.2823	0.7825	0.2853	0.7813
GH2	Num Ghosts: 10, Opacity: 0.7, Shift: 10	0.2708	0.7834	0.2721	0.7868	0.2839	0.7784
GH3	Num Ghosts: 50, Opacity: 0.5, Shift: 10, Rot: 360, Scale: 0.2	0.5259	0.4545	0.5085	0.4947	0.5165	0.5033
GH4	Num Ghosts: 20, Opacity: 0.4, Shift: 15, Rot: 180, Scale: 0.1	0.5185	0.4916	0.5391	0.4569	0.5272	0.4727
GH5	Num Ghosts: 100, Opacity: 0.6, Shift: 20, Rot: 360, Scale: 0.3	0.5057	0.5159	0.5146	0.4823	0.5043	0.5055
CA1	Shift R: 5, G: 10, B: 15	0.3140	0.7605	0.3177	0.7513	0.3188	0.7532
CA2	Shift R: 15, G: 20, B: 25	0.3188	0.7535	0.3203	0.7518	0.3160	0.7536

TABLE 4. Quantitative Comparison of PIDM Configurations (B1: Both TDB & DCF Disabled, B2: TDB Disabled, B3: DCF Disabled) Under Adversarial Attacks Against Appearance Manipulation At 256 × 176 Resolutions

Attack Type	Settings/Parameters	B1		B2		B3	
		LPIPS	SSIM	LPIPS	SSIM	LPIPS	SSIM
None	Benign	0.2374	0.7551	0.2339	0.7588	0.2393	0.7592
FP1	Mode: Salt & Pepper, Strength: 5	0.4423	0.3412	0.4576	0.3068	0.4492	0.3382
FP2	Mode: Salt & Pepper, Strength: 10	0.5502	0.1614	0.5516	0.1498	0.5332	0.2118
FP3	Mode: Salt & Pepper, Strength: 20	**0.6083**	**0.0957**	**0.6334**	**0.0755**	**0.6034**	**0.0903**
FP4	Mode: Gaussian, Strength: 5	0.2374	0.7544	0.2308	0.7584	0.2337	0.7555
FP5	Mode: Gaussian, Strength: 10	0.2422	0.7507	0.2333	0.7592	0.2364	0.7578
FP6	Mode: Gaussian, Strength: 15	0.2394	0.7529	0.2351	0.7554	0.2280	0.7597
FP7	Mode: Gaussian, Strength: 30	0.2323	0.7538	0.2340	0.7478	0.2371	0.7560
FP8	Mode: Random, Strength: 5	0.2322	0.7569	0.2297	0.7560	0.2297	0.7640
FP9	Mode: Random, Strength: 10	0.2359	0.7520	0.2365	0.7517	0.2365	0.7569
FP10	Mode: Random, Strength: 20	0.2442	0.7474	0.2324	0.7559	0.2332	0.7539
FP11	Mode: Random, Strength: 50	0.2352	0.7548	0.2284	0.7583	0.2365	0.7542
IT1	Contrast: 0.5, Brightness: 0	0.3445	0.6006	0.3526	0.5830	0.3428	0.5961
IT2	Contrast: 3.0, Brightness: 0	0.4070	0.6235	0.4107	0.6159	0.4125	0.6199
IT3	Contrast: 1.0, Brightness: -30	0.2551	0.7143	0.2626	0.7103	0.2540	0.7190
IT4	Contrast: 1.0, Brightness: 50	0.2796	0.7292	0.2870	0.7239	0.2856	0.7294
IT5	Contrast: 2.0, Brightness: 50	0.3897	0.6338	0.3842	0.6418	0.3884	0.6372
IT6	Contrast: 1.5, Brightness: 30	0.3407	0.6823	0.3433	0.6776	0.3436	0.6793
GA1	Strength: 5	0.3820	0.6592	0.3811	0.6578	0.3809	0.6515
GA2	Strength: 10	0.3869	0.6551	0.3917	0.6509	0.3947	0.6547
GA3	Strength: 20	0.3942	0.6585	0.3927	0.6561	0.3993	0.6386
GH1	Num Ghosts: 3, Opacity: 0.5, Shift: 10	0.2491	0.7581	0.2519	0.7600	0.2551	0.7593
GH2	Num Ghosts: 10, Opacity: 0.7, Shift: 10	0.2416	0.7599	0.2423	0.7633	0.2541	0.7546
GH3	Num Ghosts: 50, Opacity: 0.5, Shift: 10, Rot: 360, Scale: 0.2	0.5039	0.4133	0.4842	0.4546	0.4933	0.4628
GH4	Num Ghosts: 20, Opacity: 0.4, Shift: 15, Rot: 180, Scale: 0.1	0.4956	0.4494	0.5176	0.4137	0.5025	0.4311
GH5	Num Ghosts: 100, Opacity: 0.6, Shift: 20, Rot: 360, Scale: 0.3	0.4824	0.4759	0.4916	0.4442	0.4809	0.4666
CA1	Shift R: 5, G: 10, B: 15	0.2869	0.7396	0.2905	0.7300	0.2914	0.7323
CA2	Shift R: 15, G: 20, B: 25	0.2914	0.7323	0.2928	0.7310	0.2891	0.7326

The results in Table 3 demonstrate the effects of disabling TDB and DCF on adversarial robustness. All ablation models exhibit higher LPIPS and SSIM in comparison to PIDM during Frequency Perturbation Attacks (FP1–FP3), suggesting decreased visual fidelity. PIDM scores 0.6354 and 0.0997 for FP3 (Salt and Pepper, Strength: 20), while B1, B2, and B3 score 0.6226, 0.6496, and 0.6201 for LPIPS and 0.0987, 0.0800, and 0.0925 for SSIM, respectively. Disabling TDB (B2) considerably reduces resistance to adversarial assaults more than disabling DCF (B3), demonstrating that TDB exerts a substantial impact on model robustness.

Degradation is evident but less severe under FP Attacks (FP4-FP7), with fully functional PIDM retaining higher SSIM values and exhibiting resilience to Gaussian noise as shown in Tables 2 and 3. PIDM outperforms B1 (with both TDB

and DCF removed) in terms of brightness and contrast distortions for Intensity Transformation Attacks (IT1–IT6). GH5 exerts a substantial impact on B1, since turning off TDB and DCF makes it excessively smooth, while PIDM successfully reduces ghosting effects. The results for CA2 indicate that PIDM outperforms B1 and B2, whereas B3 (DCF disabled) only slightly enhances colour consistency, which suggests that DCF plays a role in colour preservation. Disabling TDB (B2) degrades performance, especially under FP and GH. Disabling DCF (B3) marginally improves CA but still performs worse than fully functional PIDM, as shown by additional analysis conducted at 256 $\times$ 176 resolution which can be found in Table 4. These results highlight how crucial TDB and DCF are for maintaining texture integrity and reducing adversarial distortions.

To complement the findings from the 512 $\times$ 352 resolution on DeepFashion Dataset in Table 3, the results presented in Table 4 provide a comparison of PIDM configurations under adversarial attacks at 256x176 resolution. Similar to the previous ablation study, the impact of TDB and DCF is analyzed across multiple attack types using SSIM and LPIPS as evaluation metrics. Table 1 demonstrates the impact of disabling texture diffusion block (TDB) and disentangled classifier-free (DCF) guidance on PIDM's robustness against adversarial attacks. The results demonstrate that disabling TDB (B2) leads to a more significant reduction in performance, especially under FP and GH attacks, where LPIPS values rise, and SSIM values fall compared to the fully functional PIDM. On the other hand, disabling DCF (B3) yields only marginal a matginal improvement in the case of CA, but fully functioning PIDM still performs better than B3, highlighting the importance of DCF guidance in preserving feature consistency.

Overall, PIDM achieves the highest SSIM, and the lowest LPIPS values, demonstrating its superior resilience to adversarial perturbations. However, high-intensity attacks such as FP3 and GH5 continue to cause observable degradation. Future studies should concentrate on adversarial defense techniques for image synthesis denoising diffusion models such as adaptive filtering and adversarial training to further improve the stability of PIDM under strong perturbations.

VI. DISCUSSION

This study provides significant insights into the robustness and vulnerabilities of denoising diffusion models for person image synthesis when subjected to adversarial attacks. Implementing a backdoor in a large model, such as PIDM, is both challenging and resource-intensive due to substantial differences in training costs, computational requirements, and model complexity. Our empirical findings identify ghosting and frequency-based adversarial attacks as highly effective techniques, demonstrating exceptional stealthiness, efficiency, and impact on generated outputs. By adhering to the "less is more" principle, we emphasize the simplicity and efficacy of pose-guided adversarial attacks, establishing a robust foundation for further research on adversarial injection techniques.

A. RISKS OF APPEARANCE-BASED ADVERSARIAL ATTACKS

Appearance-based adversarial attack manipulates visual attributes of the source image, such as texture, brightness, or noise, to compromise the perceptual quality of the generated outputs. These attacks exploit subtle visual features, introducing imperceptible perturbations to human observers but highly disruptive to the synthesis process. Defending against appearance-based adversarial attacks is particularly challenging in scenarios where users lack access to the underlying model. Without prior knowledge of potential triggers, manual inspection of every pixel becomes impractical. Our findings highlight frequency-based adversarial attacks as particularly effective in this context, leveraging modifications in the frequency domain to degrade output quality while maintaining imperceptibility. This underlines the need for advanced automated detection mechanisms to safeguard against such attacks, especially in real-world applications.

B. CHALLENGES IN DEFENDING POSE-CONTROLLED SYNTHESIS

Pose control attacks target the spatial configuration of pose inputs, often through IM techniques. This attack disrupts the geometric alignment of the generated outputs, leading to visually inconsistent results. While common defense strategies include verifying poses to identify potential triggers, these methods often fall short. Users are typically unaware of which pose in the dictionary serves as a "mapped pose," making it inherently difficult to detect and mitigate such attacks. Our study demonstrates that IM attacks can significantly compromise the integrity of generated outputs, highlighting a critical vulnerability in pose-guided synthesis. Developing robust countermeasures for pose-based adversarial attacks remains a challenging and essential area for future research.

C. LIMITATION

Compared to adversarial attacks, backdoor attacks on denoising diffusion models are inherently more intricate due to their prolonged training periods, resource-intensive requirements, reliance on large datasets, and the need for multiple *Graphics Processing Units* (GPUs). Despite their simplicity, adversarial attacks can significantly degrade both structural and visual fidelity of generated outputs. This comparative ease of execution makes adversarial attacks a more immediate and pressing concern, emphasizing the importance of robust adversarial defense mechanisms to maintain the reliability of these models.

D. IMPLICATIONS AND FUTURE DIRECTIONS

While developing complete defense systems remains future work, our experimental results provide critical baselines for protection mechanisms: (1) pose validity checks using joint distributions, and (2) appearance monitoring for attack-specific artifacts. These general directions are intentionally formulated to accommodate various implementation approaches while being grounded in our empirical attack thresholds. Future directions include:

- Developing hybrid detection frameworks that integrate appearance and pose-based adversarial detection for comprehensive protection.
- Exploring real-time defense mechanisms that can dynamically identify and mitigate adversarial triggers during the inference phase.
- Investigating adversarial robustness techniques specifically tailored for diffusion models to enhance their resilience against diverse attack vectors.

VII. CONCLUSION

This study highlights a critical vulnerability in state-of-the-art denoising diffusion models for person image synthesis, particularly the Pose-guided Image Synthesis Denoising Diffusion Model (PIDM). While these models excel in generating high-quality, pose-guided outputs through advanced pose and appearance control mechanisms, we demonstrate that they are susceptible to adversarial attacks that exploit these very controls. Through a comprehensive analysis, we classify adversarial attacks into two primary categories: (1) Attacks on the source image, encompassing Frequency Perturbations, Gaussian Aberrations, Ghosting, and Intensity Transformations, and (2) Attacks on the target pose, focusing on Induced Misdirection. Our results identify Frequency Perturbation attacks and Ghosting Attacks are found to be highly effective in degrading model performance while maintaining stealthiness and precision. These attacks expose vulnerabilities in the generated images' structural integrity and visual fidelity, highlighting the need for robust defensive measures. This work significantly contributes by systematically evaluating the PIDM model under adversarial attacks and comparing its performance against state-of-the-art methods. The findings underscore the effectiveness of the proposed attack strategies, particularly frequency-based attacks, due to their perceptual deceptiveness, efficiency, and low barrier to implementation. By leveraging a state-of-the-art diffusion model, our study lays a strong foundation for future research to develop robust defense mechanisms to enhance the security and reliability of person image synthesis models. Future work will focus on exploring adaptive adversarial detection and mitigation strategies, inspired by existing methodologies, to safeguard against such vulnerabilities. Additionally, extending the scope of this research to other generative models and exploring their adversarial resilience will be critical to ensuring the safe deployment of these technologies in real-world applications.

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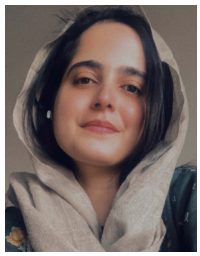
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